ASSESSING THE IMPACT OF IMPORTS OF CRITICAL MINERALS TO THE UNITED STATES

DATA 608 Project7

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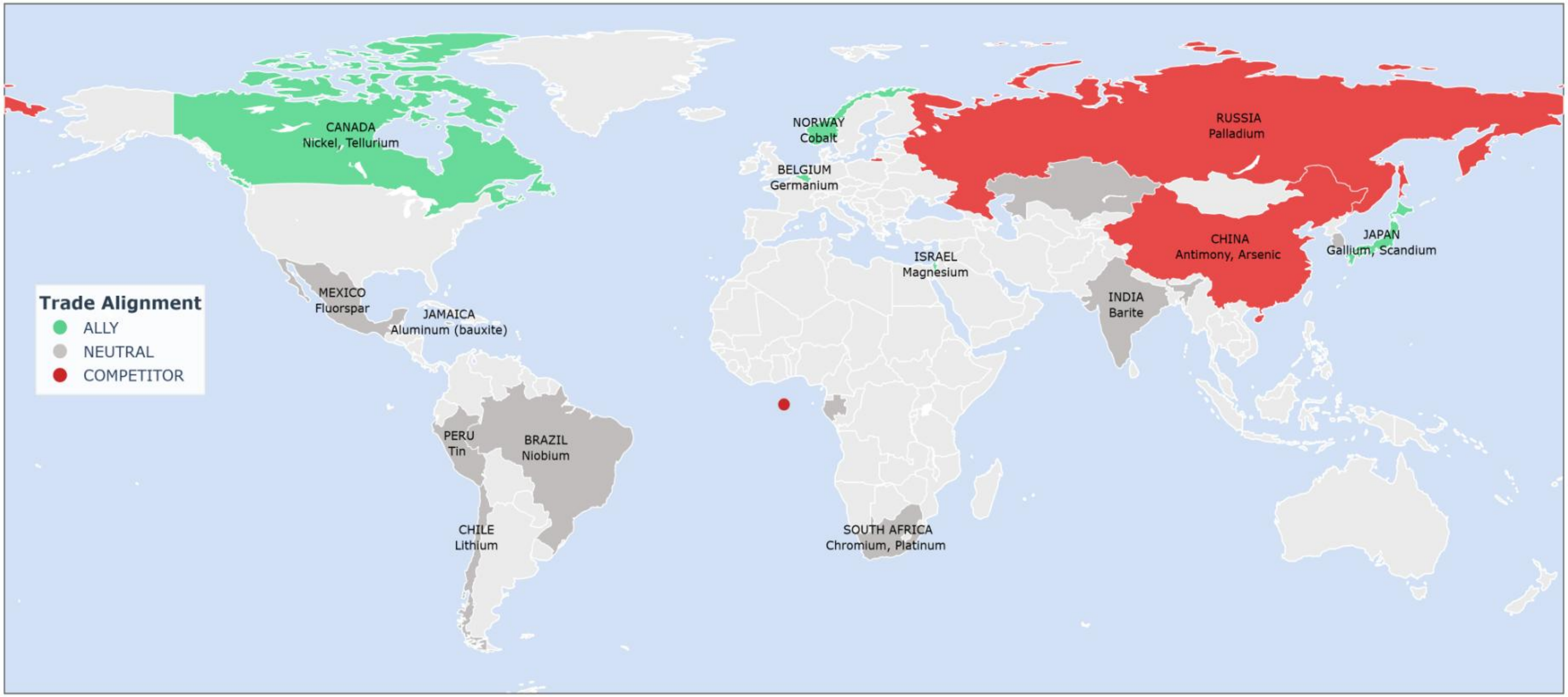
How Much of the U.S. Economy Depends on Imported Minerals?

Electronics	Energy Storage	Industrial Metals
71%	**75%**	**58%**
≈ $448B Industry Share of GDP	≈ $366B Industry Share of GDP	≈ $345B Industry Share of GDP

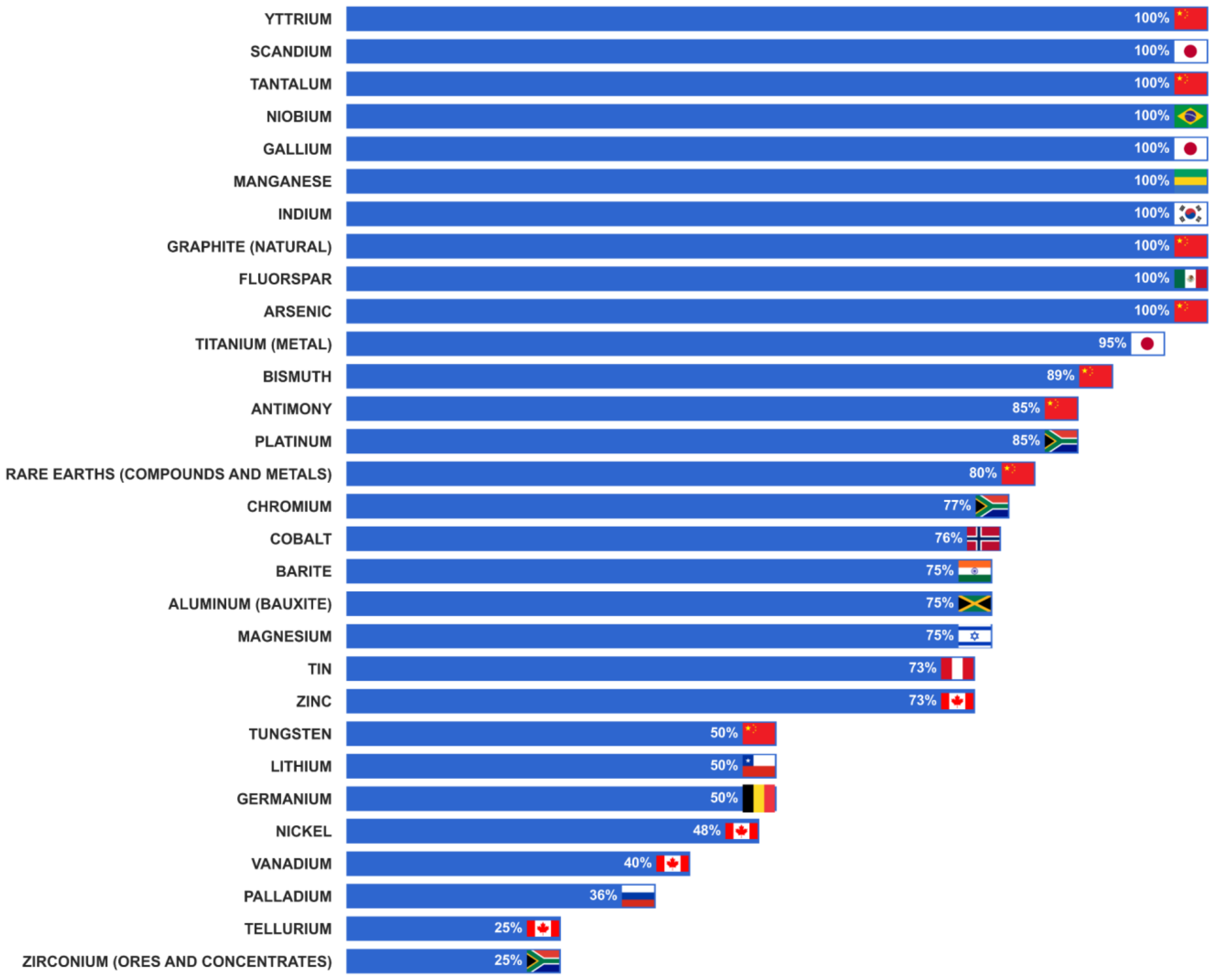
- This slide focuses on the GDP weight behind each category.
- ***Electronics is a $448B sector - high reliance means high economic risk*.**
- ***Energy storage has the highest reliance (75%), which basically confirms that EV supply chains are vulnerable*.**
- ***Industrial metals are in the middle, but still meaningful because they support manufacturing overall*.**

U.S. Critical Mineral Import Partners by Trade Alignment (Top 2 Minerals per Country, 2024)



- Now the geopolitical angle: not all suppliers are equal.
- Problem: a lot of critical minerals come from neutral or competitor countries.
- This slide makes it obvious that the supply chain is not aligned with U.S. strategic interests.

America's Import Reliance on Critical Minerals
Country of Highest Dependence



Full Import Reliance

100%

Graphite, Manganese, Rare Earths

High-Risk Minerals

>75%

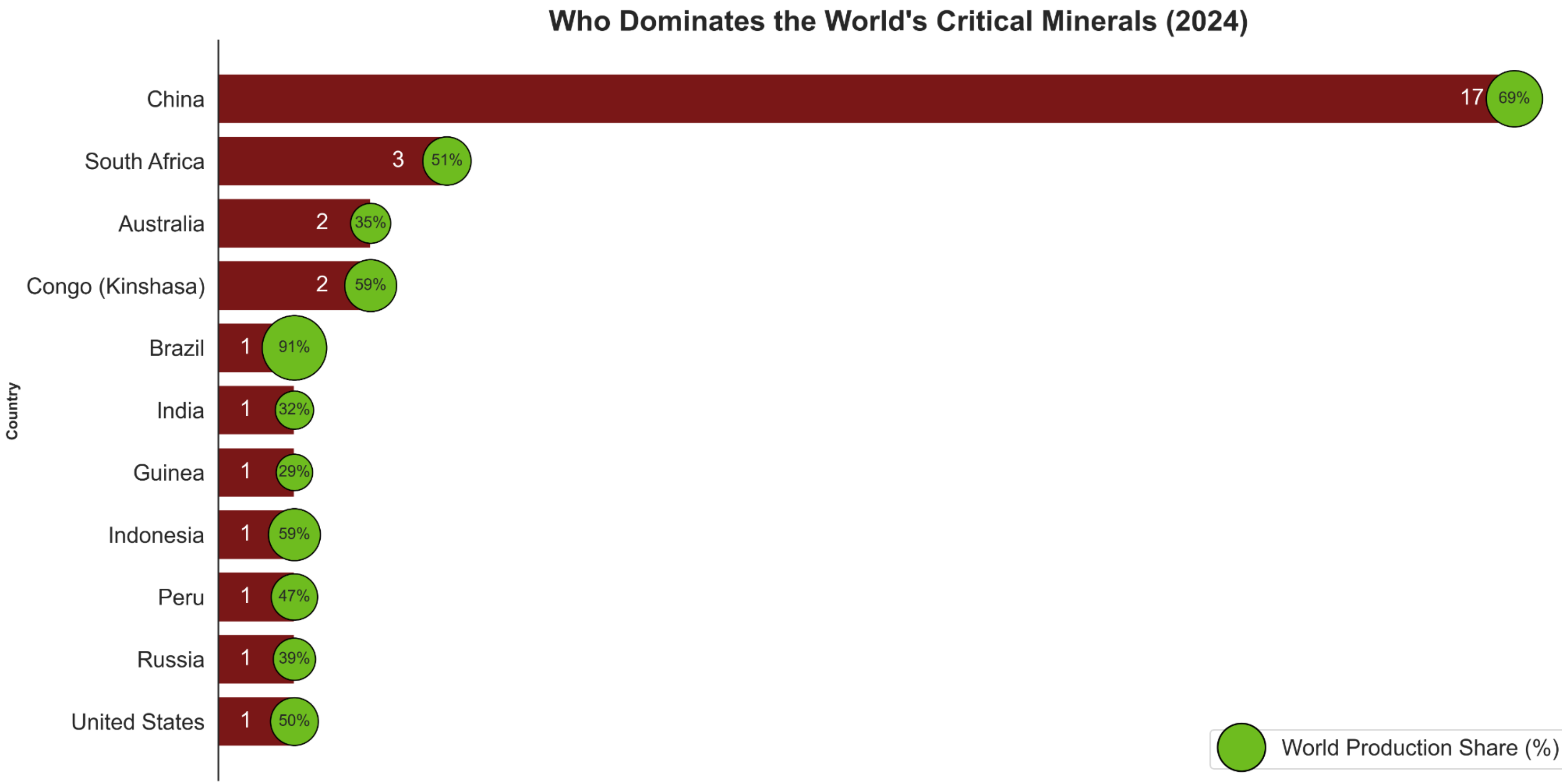
20+ minerals used across industries

Primary Supplier

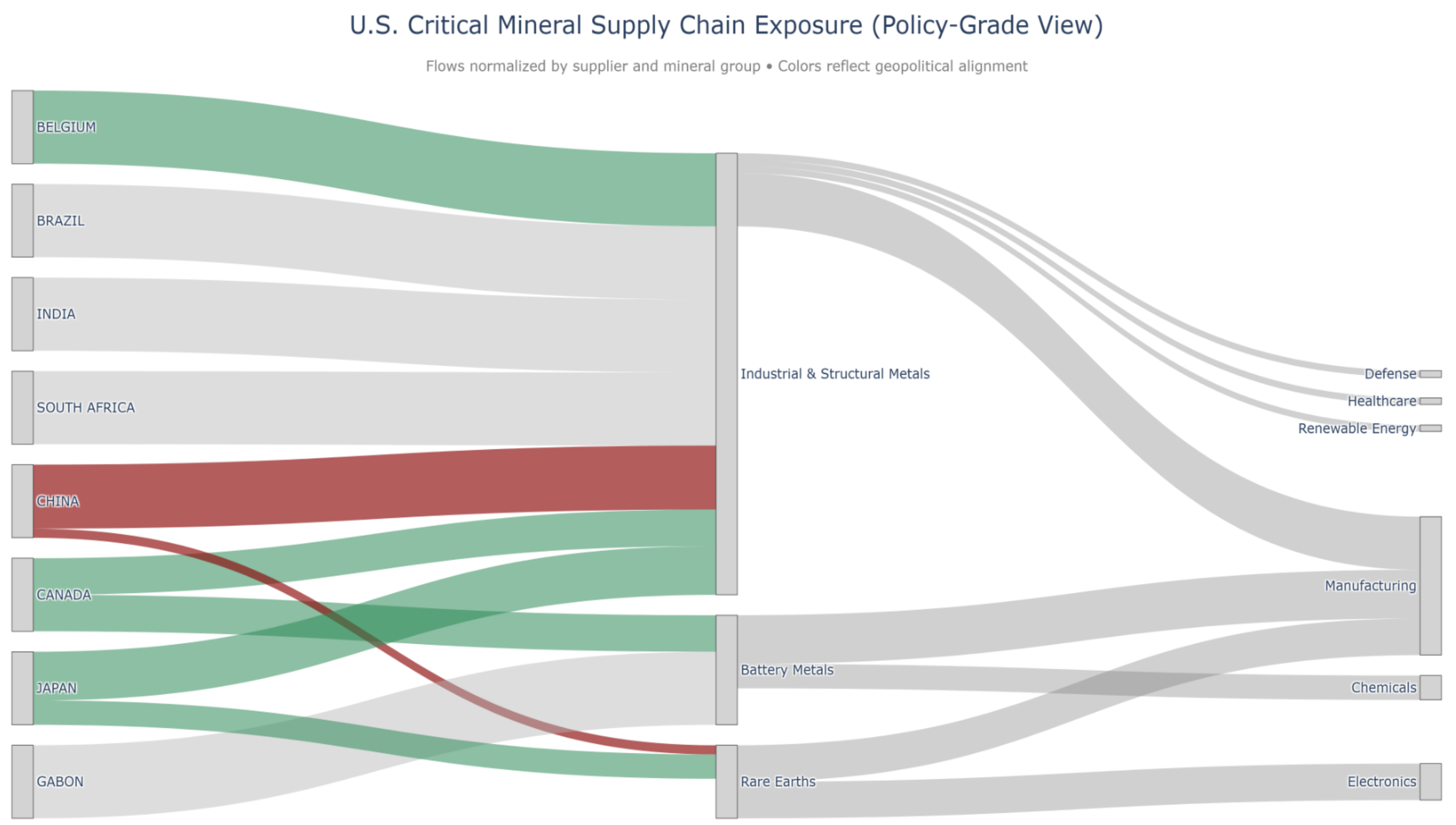
China

Largest source for battery & Rare Earth inputs

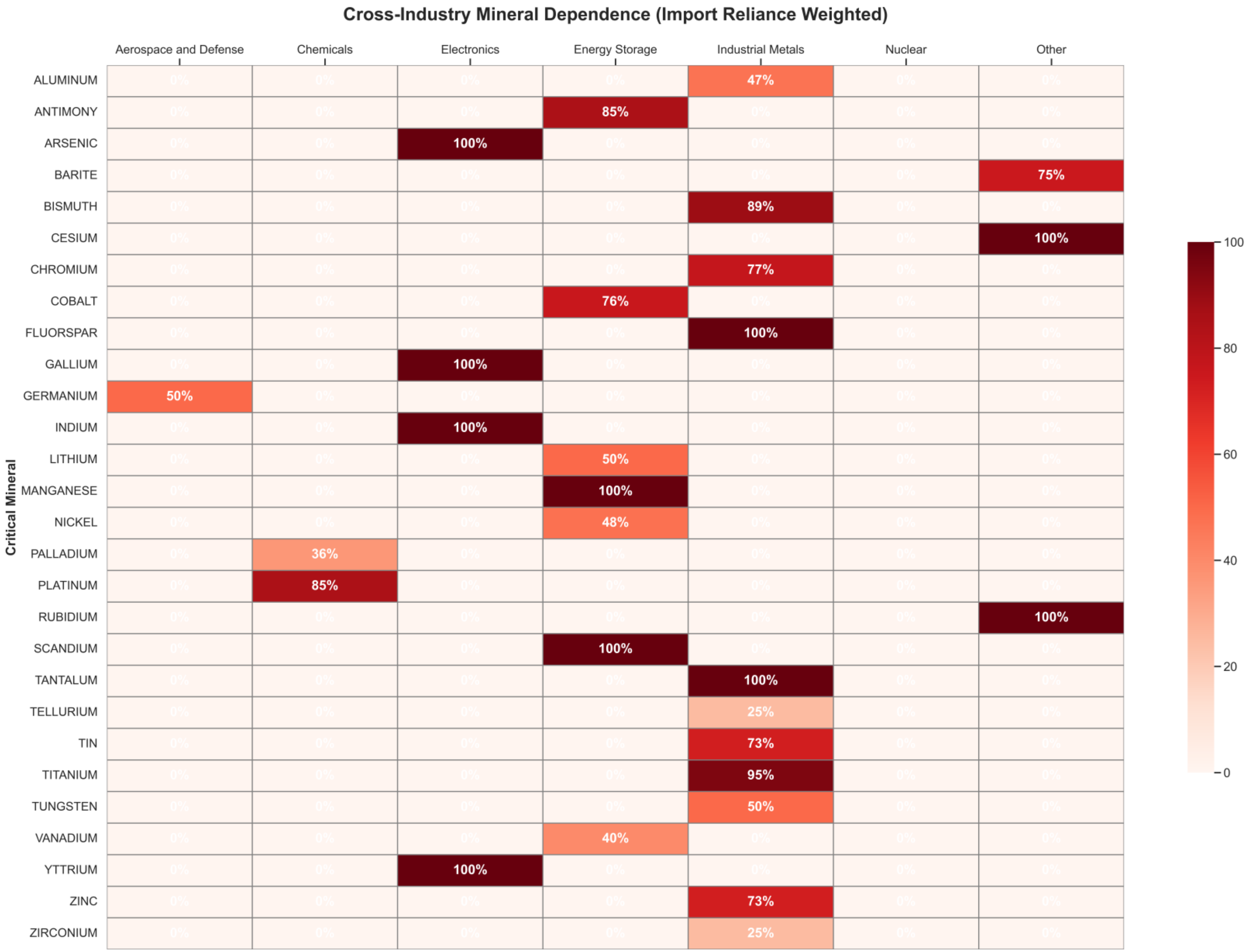
- Here's the mineral-by-mineral breakdown.
- Most of the top dependency ***minerals are 75–100% imported***.
- What matters is not just the percentage but *who* the U.S. depends on - ***China dominates a lot of these bars***.
- For some minerals (palladium, chromium, cobalt), we rely on countries with either unstable production or geopolitical tension.



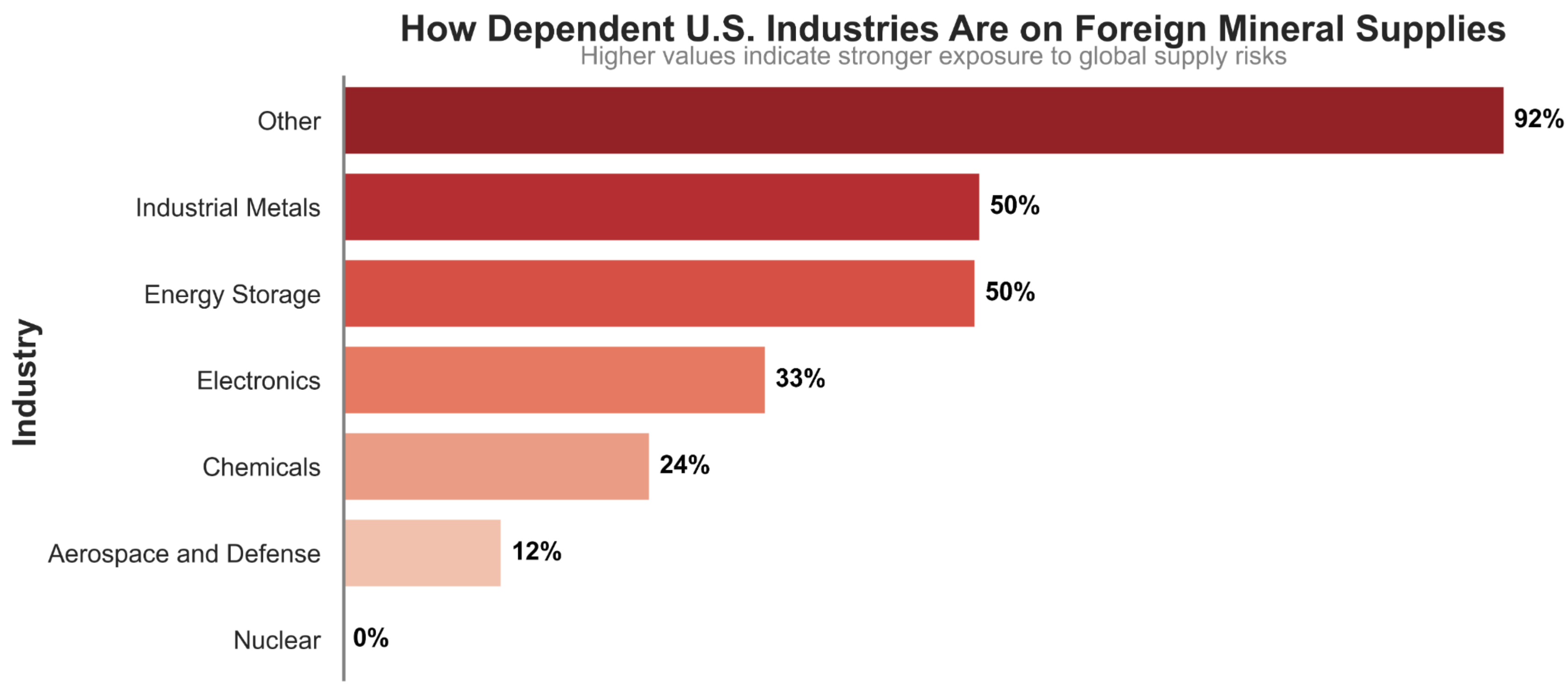
- This chart answers the question: “Who controls global production?”
- China is in its own league - dominating 17 minerals and nearly 70% average global share.
- A few countries like South Africa, Congo, Brazil also show strong dominance in specific minerals.
- This matters because even if the U.S. doesn’t directly import from these countries, *the global price and supply chain are influenced by them*



- This Sankey chart visualizes upstream → midstream → downstream flow.
- Industrial metals and battery metals dominate both volume and strategic importance.
- You can see how a few countries feed almost every downstream sector.
- The key point: a disruption at the supplier level cascades into manufacturing, chemicals, and electronics quickly.



- This heatmap connects specific minerals to industries.
- We can immediately see where "single points of failure" exist — for example, cobalt and lithium lighting up energy storage.
- Electronics heavily depends on gallium, indium, germanium — all almost fully imported.
- The more red an industry is, the fewer domestic options it really has.



- Instead of thinking only in minerals, here's the industry translation.
- *Industrial metals, Energy storage and Electronics* are the most exposed.
- If supply is disrupted, these sectors would feel it first — anything from EV production delays to higher semiconductor prices.
- Nuclear is the only sector that looks insulated here (pretty rare to see a clean zero).

America's Mineral Supply Chain Depends Heavily on Imports

Value at Risk

$1.9T

GDP in exposed sectors

Import Reliance

67%

Avg. reliance across key sectors

Critical Materials

20+

>75% import dependent

- So the picture is pretty simple: **$1.9T in exposed value, 67% average import reliance, and 20+ minerals where we're wide open.**
- And most of that exposure runs through countries that aren't exactly aligned with U.S. interests. That's the real concern - the leverage it creates.
- Solving this isn't about replacing every import. It's about:
  - **Reduce single-country dominance – or at least build more stable trade relations when we can (especially with China)**
  - **Building U.S. capacity where it actually pays off.**
  - **Leaning more on trusted partners.**
- We won't eliminate dependency, but we *can* reshape it - so strategic competitors have less control over critical U.S. industries.